Cpts 317. 8/29.

Last time:

$L_1 \cup L_2$

$L_1 \cdot L_2$

Today:

1. $L^*$ Kleene star

2. Def. of Regular languages.

$$L^* = \bigcup_{k \geq 0} L^k$$

$$L^k = \underbrace{L \cdot L \cdot \ldots \cdot L}_{k \text{ times.}}$$

$$= L^0 \cup L^1 \cup L^2 \cup \ldots\ldots$$

$$L^+ = \bigcup_{k \geq 1} L^k$$

$$= L^1 \cup L^2 \cup L^3 \cup \ldots\ldots$$

Special cases:

$L^0 = \{\Lambda\}$ even when $L = \emptyset$;

$\emptyset^0 = \{\Lambda\}, \quad \emptyset^1 = \emptyset = \emptyset^2 = \ldots$

$\{\Lambda\}^0 = \{\Lambda\}, \quad \{\Lambda\}^1 = \{\Lambda\} = \{\Lambda\}^2 = \ldots$

Consequently, $\{\Lambda\}^* = \{\Lambda\}, \quad \{\Lambda\}^+ = \{\Lambda\}$

$\emptyset^* = \{\Lambda\}, \quad \emptyset^+ = \emptyset.$

$L = \{a, ab\}^*$

(1). Describe L in English:

L is formed by, exactly, the following words: concatenating for any (including 0) # of times a word picked from $\{a, ab\}$.

(2). Can you see:

$\Lambda \in L$.

$aab \in L$.

$aba \in L$.

$\underline{ab}\,\underline{a}\,\underline{ab} \in L$

$abb \notin L$

$aaab \in L$.

L is an inf. language.

What's the diff between

$$\{a, ab\}^* \quad \text{and} \quad \{a, ab\}^+$$

ans: $\{a, ab\}^* = \{a, ab\}^+ \cup \{\lambda\}$.

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Example. $L = \{01, 110\}^*$.

Example in $L$: $1100101 \in L$

counter Example in $L$: $1100 \notin L$.

Regular languges. ( is a class of languges).

$\uparrow \mathcal{R}$

you can't say "set" here.

1. $\emptyset \in \mathcal{R}$

2. $\{\Lambda\} \in \mathcal{R}$

3. $\{a\} \in \mathcal{R}, \quad a \in \Sigma$.

4. if $L_1, L_2 \in \mathcal{R}$, Then

   4.1. $L_1 \cup L_2 \in \mathcal{R}$

   4.2. $L_1 \cdot L_2 \in \mathcal{R}$

   4.3. $L_1^* \in \mathcal{R}$.

Examples. $\Sigma = \{a, b, c\}$.

(1). $\{a, b\}^*$ is a reg language. why?

$\{a\}$ and $\{b\}$ are reg languges —— 3

$\{a, b\}$ is reg language —— 4.1

$\{a, b\}^*$ is reg language —— 4.3

(2). $\{ccc\}$ is reg language. why?

$\{c\}$ is reg —— 3

$\{cc\}$ is reg —— 4.2

$\{ccc\}$ is reg —— 4.2

(3). $\{a,b\}^* \{ccc\}$ is reg. why?

using 4.2 on $L_1 = \{a,b\}^*$ in (1) and

$L_2 = \{ccc\}$ in (2).

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Regular Expressions:

(a regExp rep. a reg language) — a set of strings.

(a regExp doesn't rep. a string)

regExp's are defined as:

(1). $\emptyset$ is a regExp

(2). $\Lambda$ is a regExp

(3). $a$ is a regExp, $a \in \Sigma$.

(4). if $r_1$, $r_2$ are regExp, then so are:

(4.1). $r_1 + r_2$

(4.2). $r_1 r_2$

(4.3). $r_1^*$

Examples:

$$(a + b)^*$$

loop

nondet. choice $\equiv$ "OR".

all strings on $\Sigma = \{a, b\}$.

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Key:

$*$ —— loop for 0 or more times

$+$ —— nondet. choice

$\cdot$ —— Concatenation.
(one is before another).

RegExp for traffic light:

GYR GYR .....

answer:

$$(GYR)^*(\Lambda + G + GY)$$

Alice shopping Exmple.

Alice lives at Pullman.

does some shoping at Moscow, comes back,

Then goes to Moscow again, .....

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pullman, moscow, pullman.

<u>pullman, mosca</u>, <u>pullam, moscow</u>, pullman.

⋮

$(\text{pullman moscow})^*$ pallman

Elevator:

floor3.

floor2.

floor1.,

You take the elevator for floor1 and, finally, you come back to floor1.

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I want: RegExp for all the sequences of floors you travel.